BIOCLIMATIC BUILDING DESIGN 2024-25

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Report on Thermal Analysis in Winter, Summer

SAPIENZA
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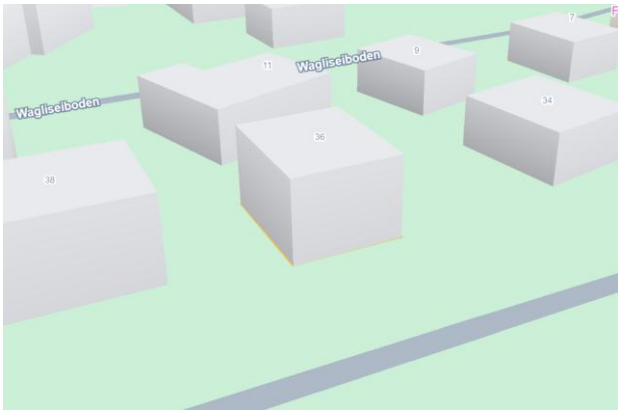
1. Site and Building Description

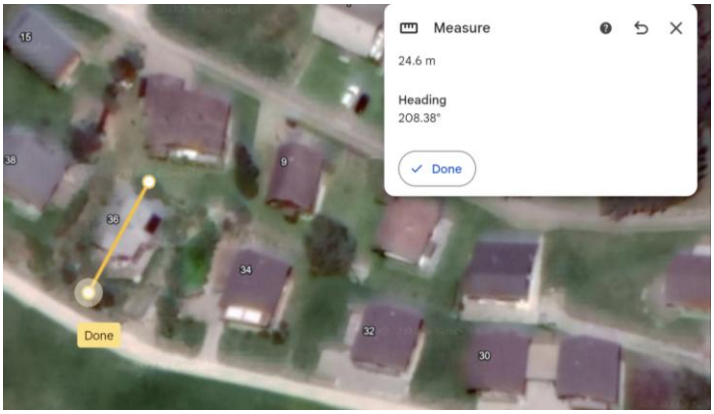
The case study analyzes a residential chalet located in Flühli, canton Luzern, Switzerland. Situated at an elevation of 1322 meters, the site experiences alpine climatic conditions with strong seasonal contrasts and variable solar exposure.

Location and Orientation: - Address: Bergblick, Wagliseiboden 13, 6174 Flühli, Switzerland - Coordinates: 46.81502° N, 7.98991° E - Orientation of main glazed façade: 208.38° (south-west)

Geometry Simplification: To facilitate modeling, the building was simplified to a rectangular prism: - Length: 12.77 m, Width: 10.70 m, Height: 7.50 m

This approximation allows consistent surface area calculations across walls, floor, and ceiling.

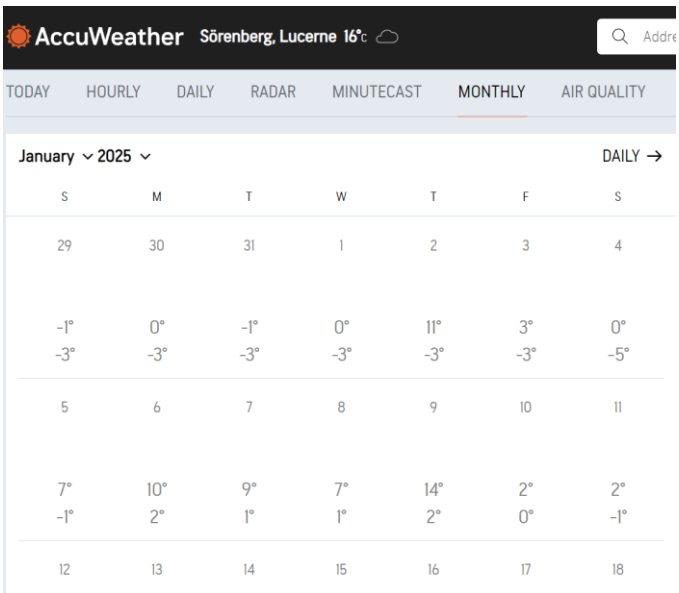


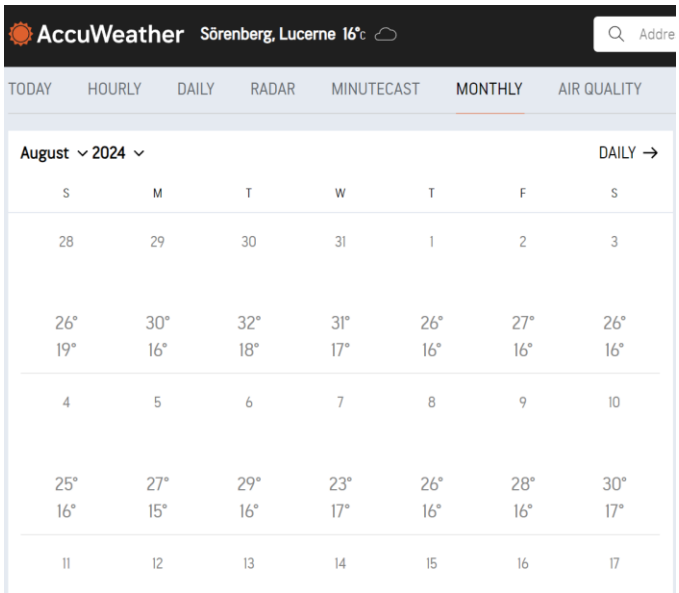


2. Climatic Conditions

Summer Simulation: - Date: 15 August 2024, 12:00 - Outdoor temperature: max 26 °C - Indoor setpoint: 23 °C - Solar irradiance assessed using PVGIS and SunCalc

Winter Simulation: - Date: 17 January 2025, 12:00 - Outdoor temperature: 0 °C to 2 °C (1 °C used) - Indoor setpoint: 20 °C



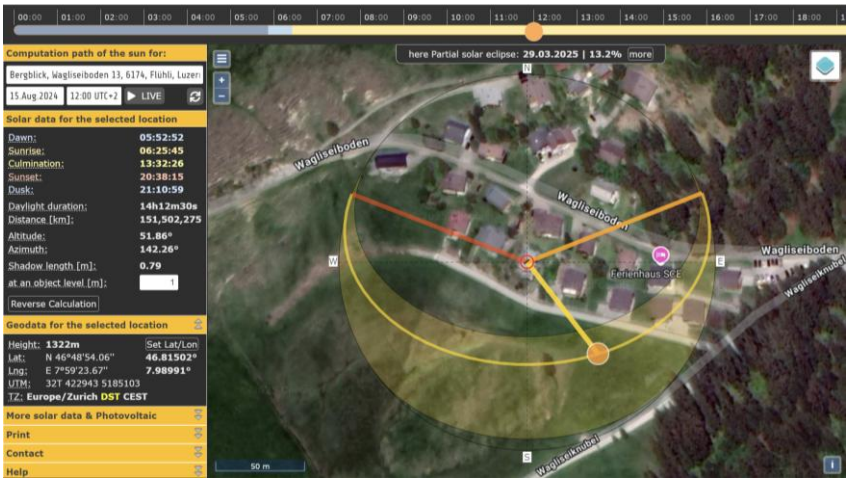


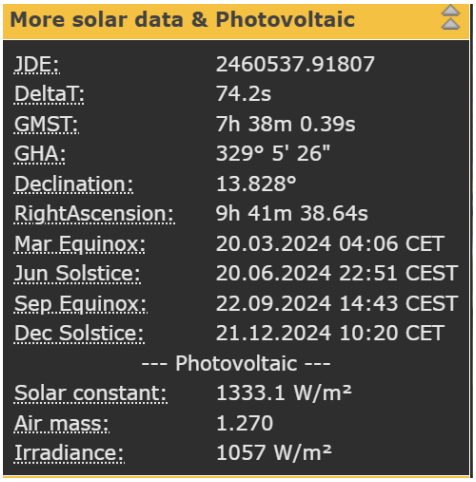
3. Solar Irradiance Calculation

PVGIS provided monthly average irradiance (~1004 W/m²) for SW vertical surfaces. However, for accuracy at a specific hour, **SunCalc** data was used: - Sun azimuth: 142.26° - SW wall azimuth: 208.38° - Angle of incidence: 66.12° (determines effective solar input relative to surface orientation) - Horizontal irradiance: 1057 W/m² - Resulting projected irradiance on SW wall:

$$I_{SW} = 1057 \cdot \cos(66.12^\circ) \approx \boxed{428.1\ \mathrm{W/m^2}}$$

This value was used for solar gain calculations due to its time-specific accuracy.



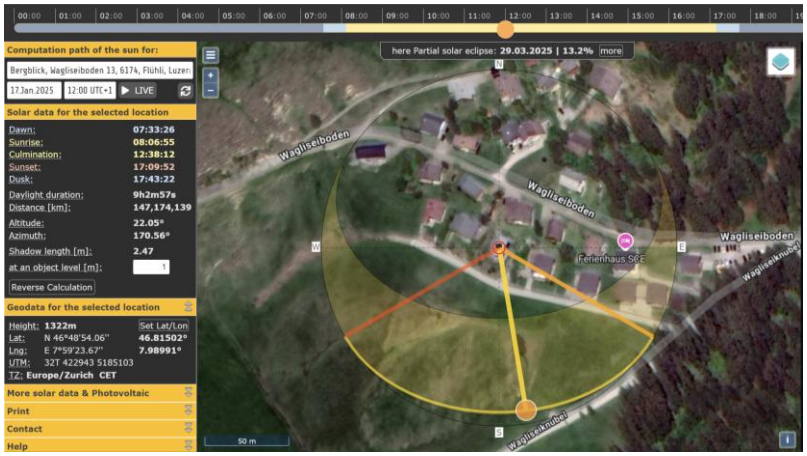


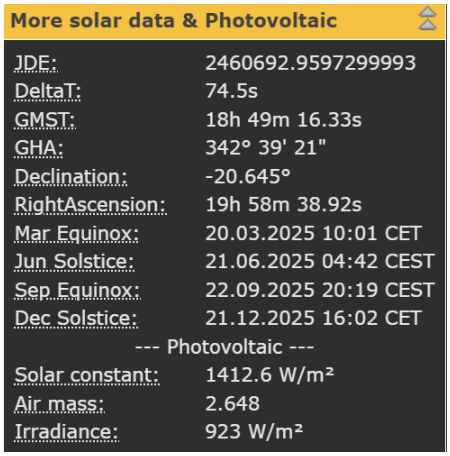
4. Exposition Correction Factor f_x

The exposition correction factor f_x is applied to heat transfer calculations in winter to account for the varying levels of exposure different building surfaces have to the environment. This includes factors such as

orientation, terrain shielding, and wind exposure. Values were adapted from standard tables and verified against solar geometry:

Surface	Orientation	f_x	Rationale
SW Wall	208.38°	0.51	Window-dominated, terrain shielding
SE Wall	~132°	0.54	Solid construction, less shielding
NE Wall	~47°	1.05	Cold-facing, shaded
NW Wall	~317°	1.02	Less sun, slightly more exposed than NE
Floor	Horizontal	1.00	Over crawlspace
Ceiling	Horizontal	1.10	Wind-exposed roof





5. Envelope U-Values

Component	Description	U (W/m²·K)	Justification
Walls	Timber + mineral wool	0.22	Based on ISO 6946 and passive design norms
Windows	Double-glazed, wood/aluminum frame	1.20	Typical of modern residential low-E windows
Floor	Raised timber floor over crawlspace	0.28	Includes insulation; mild external contact
Ceiling	Sloped insulated roof	0.18	Reflects thick rafter insulation layer

6. Air Exchange Rate

A value of **0.8 ACH** was used in winter to model natural infiltration and manual ventilation: - Matches ASHRAE and SIA guidance for non-passive homes - Accounts for wind-driven infiltration in alpine winter

This contributes to overall heat loss via the following equation:

$$\dot{Q}_{\text{air}} = n \cdot V \cdot f \cdot \rho \cdot c_p \cdot \Delta T$$

Where: - $n = 0.8$ ACH, $f = 1/3600$, $\rho = 1.25$, $c_p = 1000$

7. Equivalent Temperature Difference (ΔT_{eq})

In the summer model, ΔT_{eq} was used for opaque surfaces to reflect solar and convective gains:

Surface	ΔT_{eq} (°C)	Basis
SW Wall	7.5	High solar exposure
SE Wall	5.0	Morning sun
NE Wall	1.5	Mostly shaded
NW Wall	1.0	Low lateral exposure
Ceiling	10.0	Full sun, roof surface

Derived from EN ISO 13790, adjusted using SunCalc data and material absorptivity.

8. Glazing Transmittance τ

For solar gain through windows, a transmittance of $\tau = 0.675$ was used: - Based on average of double-glazed low-E windows - Adjusted for **66.12° angle of incidence** using cosine law - Reflects mix of clarity and frame material (wood/aluminum)

9. Internal Gains

9.1 People - Occupancy: 3 people - Heat gain: Sensible = 72.5 W, Latent = 52.5 W per person - Chosen based on average activity level (between seated and light movement), per ASHRAE and ISO 7730 - Total: $3 \cdot (72.5 + 52.5) = 375$ W

9.2 Lighting - Type: Mixed LED + halogen - Power density: $q_{\text{light}} = 12.5\ \text{W/m}^2$ - Represents midpoint for semi-efficient lighting systems, consistent with mixed-use residential lighting

10. Summary of Applied Formulations

Envelope Heat Transfer: - Opaque Surfaces (Winter):

$$\dot{Q} = U \cdot A \cdot \Delta T$$

- Opaque Surfaces (Summer):

$$\dot{Q} = U \cdot A \cdot \Delta T_{eq}$$

Window Heat Transfer and Solar Gain:

$$\dot{Q}_{\text{window}} = U \cdot A \cdot \Delta T + I \cdot A \cdot \tau$$

Air Exchange Losses:

$$\dot{Q}_{\text{air}} = n \cdot V \cdot \frac{1}{3600} \cdot 1.25 \cdot 1000 \cdot \Delta T$$

Internal Gains: - People:

$$\dot{Q}_{\text{people}} = n \cdot (Q_{\text{sens}} + Q_{\text{lat}})$$

- Lighting:

$$\dot{Q}_{\text{light}} = q_{\text{light}} \cdot A$$